

A Just and Comprehensive Strategy for Using NLP to Address Online Abuse

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Abstract

Online abusive behavior affects millions and the NLP community has attempted to mitigate this problem by developing technologies to detect abuse. However, current methods have largely focused on a narrow definition of abuse to detriment of victims who seek both validation and solutions. In this position paper, we argue that the community needs to make three substantive changes: (1) expanding our scope of problems to tackle both more subtle and more serious forms of abuse, (2) developing proactive technologies that counter or inhibit abuse before it harms, and (3) reframing our effort within a framework of justice to promote healthy communities.

1 Introduction

Online platforms have the potential to enable substantial, prolonged, and productive engagement for many people. Yet, the lived reality on social media platforms falls far short of this potential (Papacharissi, 2004). In particular, the promise of social media has been hindered by antisocial, abusive behaviors such as harassment, hate speech, trolling, and the like. Recent surveys indicate that abuse happens much more frequently than many people suspect (40% of Internet users report being the subject of online abuse at some point), and members of underrepresented groups are targeted even more often (Herring et al., 2002; Drake, 2014; Anti-Defamation League, 2019).

The NLP community has responded by developing technologies to identify certain types of abuse and facilitating automatic or computer-assisted content moderation. Current technology has primarily focused on overt forms of abusive language and hate speech, without considering both (i) the success and failure of technology beyond getting the classification correct, and (ii) the myriad forms that abuse can take.

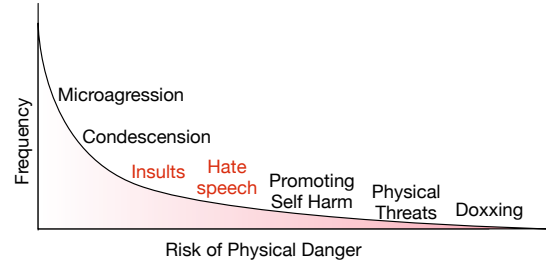


Figure 1: Abusive behavior online falls along a spectrum, and current approaches focus only on a narrow range (shown in red text), ignoring nearby problems. Impact comes from both the frequency (on left) and real-world consequences (on right) of behaviors. This figure illustrates the spectrum of online abuse in an hypothetical manner, with its non-exhaustive examples inspired from prior surveys of online experiences (Dugan, 2017; Salminen et al., 2018).

As Figure 1 shows, a large spectrum of abusive behavior exists—some with life-threatening consequences—much of which is currently unaddressed by language technologies. Explicitly hateful speech is just one tool of hate, and related tactics such as rape threats, gaslighting, First Amendment panic, and veiled insults are effectively employed both off- and online to silence, scare, and exclude participants from what should be inclusive, productive discussions (Filipovic, 2007).

In this position paper, we argue that to promote healthy online communities, three changes are needed. First, the NLP community needs to rethink and expand what constitutes abuse. Second, current methods are almost entirely *reactive* to abuse, entailing that harm occurs. Instead, the community needs to develop proactive technologies that assist authors, moderators, and platform owners in preventing abuse before it occurs. Finally, we argue that both of these threads point to a need for a broad re-aligning of our community goals towards *justice*, rather than simply the elim-

ination of abusive behavior. In arguing for these changes, we outline how each effort offers new challenging NLP tasks that have concrete benefits.

2 Rethinking What Constitutes Abuse

The classifications we adopt and computationally enforce have real and lasting consequences by defining both what is and what is not abuse (Bowker and Star, 2000). Abusive behavior is an omnibus term that often includes harassment, threats, racial slurs, sexism, unexpected pornographic content, and insults—all of which can be directed at other users or at whole communities (Davidson et al., 2017; Nobata et al., 2016). However, NLP has largely considered a far narrower scope of what constitutes abuse through its selection of which types of behavior to recognize (Waseem et al., 2017; Schmidt and Wiegand, 2017; Fortuna and Nunes, 2018). We argue that NLP needs to expand its computational efforts to recognize two additional general types of abuse: (a) infrequent and physically dangerous abuse, and (b) more common but subtle abuse. Additionally, we need to develop methods that respect community norms in classification decisions. These categories of abuse and the importance of community norms have been noted elsewhere (Liu et al., 2018; Guberman and Hemphill, 2017; Salminen et al., 2018; Blackwell et al., 2017) but have not yet received the same level of attention in NLP.

Who has a right to speak and in what manner are subjective decisions that are guided by social relationships (Foucault, 1972; Noble, 2018), and the specific choices our algorithms make about what speech to allow and what to silence have powerful effects. For instance, rejecting behavior as not being abusive because it is outside the scope of our classification can cause substantial harm to victims (Blackwell et al., 2017), tacitly involving the NLP community in algorithmic bias that sanctions certain forms of abuse. Thus, categorization is particularly thorny: a broad categorization is likely too computationally inefficient, yet a narrow categorization risks further marginalizing affected community members and can lead to lasting harm. Following, we outline three key directions for the community to expand its definitions.

2.1 Physically Threatening Online Abuse

We outline three computational challenges related to infrequent but overt physically-manifesting

abuse that NLP could be applied to solve. First, such behaviors do not necessarily adopt the language of hate speech or more common forms of hate speech and may in some contexts appear innocuous but are clearly dangerous in others. For example, posting a phone number to call could be acceptable if one is encouraging others to call their political representative, yet would be a serious breach of privacy (doxxing) if posted as part of a public harassment campaign. Similarly, declarations of “keep up the weight loss!” may be positive in a dieting community, yet reinforce dangerous behavior in a pro-anorexia community. Speech that in isolation appears offensive, such as impoliteness or racial slurs, may serve pro-social functions such as promoting intimacy (Culpeper, 1996) or showing camaraderie (Allan, 2015).

Second, behaviors such as swatting, human trafficking, or pedophilia have all occurred on public social media platforms (Jaffe, 2016; Latonero, 2011; Holt et al., 2010). However, methods have yet to be developed for recognizing when users are engaging in these behaviors, which may involve coded language, and require recognizing these alternative forms. Current approaches for learning new explicitly-hateful symbols could be adapted to this task (e.g., Roy, 2016; Gao et al., 2017).

Third, online platforms have been used to incite mobs of people to violence (Siegel, 2015). These efforts often use incendiary fake news that plays upon factional rivalries (Samory and Mitra, 2018). Abusive language detection methods can build upon recent advances at detecting fake news to identify content-sharing likely to lead to violence (McLaughlin, 2018; Oshikawa et al., 2018).

2.2 Subtle Abuse

Many forms of abusive behavior are linguistically subtle and implicit. Behaviors such as condescension, minimization (e.g., “your situation isn’t that bad”), benevolent stereotyping, and microaggressions are frequently experienced by members of minority social groups (Sue et al., 2007; Glick and Fiske, 2001). While subtle, such abuse can still be as emotionally harmful as overt abuse to some individuals (Sue, 2010; Nadal et al., 2014). The NLP community has two clear paths for growth into this area.

First, although recognized within the larger NLP abuse typology (Waseem et al., 2017), only a handful of approaches have attempted these prob-

lems, such as identifying benevolent sexism (Jha and Mamidi, 2017), and new methods must be developed to identify the implicit signals. Successful approaches will likely require advances in natural language understanding, as the abuse requires reasoning about the implications of the propositions. A notable example of such an approach is Dinakar et al. (2012) who extract implicit assumptions in statements and use common sense reasoning to identify social norm violations that would be considered insults.

Second, new methods should identify *disparity* in treatment of social groups. For example, in a study of the respectfulness of police language, Voigt et al. (2017) found that officers were consistently less likely to use respectful language with black community members than with white community members—a disparity in a *positive* social dimension. As NLP solutions have been developed for other social dimensions of language such as politeness (Danescu-Niculescu-Mizil et al., 2013; Munkova et al., 2013; Chhaya et al., 2018) and formality (Brooke et al., 2010; Sheikh and Inkpen, 2011; Pavlick and Tetreault, 2016), these methods could be readily adapted for identifying such systematic bias for additional social categories and settings.

2.3 Community Norms Need to be Respected

Social norms are rules and standards that are understood by members of a group, and that guide and constrain social behavior without the force of laws (Triandis, 1994; Cialdini and Trost, 1998). Norms can be nested, in that they can be adopted from the general social context (e.g., use of pejorative adjectives are rude), and more general internet comment etiquette (e.g., using all caps is equivalent to shouting). Yet, norms for what is considered acceptable can vary significantly from one community to another, making it challenging to build one abuse detection system that works for all communities (Chandrasekharan et al., 2018).

Current NLP methods are largely context- and norm-agnostic, which leads to situations where content is removed unnecessarily when deemed inappropriate (i.e., false positives), eroding community trust in the use of computational tools to assist in moderation. A common failure mode for sociotechnical interventions like automated moderation is failing to understand the online community where they are being deployed (Krishna,

2018). Such community-specific norms and context are important to take into account, as NLP researchers are doubling down on context-sensitive approaches to define (e.g., Chandrasekharan and Gilbert, 2019) and detect abuse (e.g., Gao and Huang, 2017).

However, not all community norms are socially acceptable within the broader world. Even behavior considered harmful in one community might be celebrated in another, e.g., Reddit’s r/fatpeoplehate (Chandrasekharan et al., 2017), and Something Awful Forums (Pater et al., 2014). The existence of problematic normative behaviors within certain atypical online communities poses a challenge to abuse detection systems. Fraser (1990) notes that when a public space is governed by a dominant group, its norms about participation end up perpetuating inequalities. One approach to address this challenge would be to work closely with the different stakeholders involved in online governance, like platform administrators, policy makers, users and moderators. This will enable the development of solutions that cater to a wider range of expectations around moderating abusive behaviors on the platform, especially when dealing with deviant communities.

2.4 Challenges for Creating New NLP Shared Tasks on Abusive Behavior

Shared tasks have long been an NLP tradition for establishing evaluating metrics, defining data guidelines, and, more broadly, bringing together researchers. The broad nature of abusive behavior creates significant challenges for the shared task paradigm. Here, we outline three opportunities for new shared tasks in this area. First, new NLP shared tasks should develop annotation guidelines accurately define what constitutes abusive behavior in the target community. Recent works have begun to make progress in this area by modeling the context in which a comment is made through user and community-level features (Qian et al., 2018; Mishra et al., 2018; Ribeiro et al., 2018), yet often the norms in these settings are implicit making it difficult to transfer the techniques and models to other settings. As one potential solution, Chandrasekharan et al. (2018) studied community norms on Reddit in a large-scale, data-driven manner, and released a dataset of over 40K removed comments from Reddit labeled according to the specific type of *norm* being violated (Chan-

drasekharan and Gilbert, 2019).

Second, new NLP shared tasks must address the data scarcity faced by abuse detection research while minimizing harm caused by the data. Constant exposure to abusive content has been found to negatively and substantially affect the mental health of moderators and users (Roberts, 2014; Gillespie, 2018; Saha et al., 2019). However, labeled ground truth data for building and evaluating classifiers is hard to obtain because platforms typically do not share moderated content due to privacy, ethical and public relations concerns. One possibility for significant progress is to work with platform administrators and stakeholders to make proprietary data available as private test sets on platforms like Codalab, thereby keeping annotations in line with community norms and still allowing researchers to evaluate on real behavior.

Third, tasks must clearly define *who* is the end-user of the classification labels. For example, will moderators use the system to triage abusive content, or is the goal to automatically remove abusive content? Current solutions are often trained and evaluated in a *static* manner, only using pre-existing data; whether these solutions are effective upon deployment remains relatively unexplored. Evaluation must go beyond just traditional measures of performance like precision and recall, and instead begin optimizing for metrics like reduction in moderator effort, speed of response, targeted recall for severe types of abuse, moderator trust and fairness in predictions.

3 Proactive Approaches for Abuse

Existing computational approaches to handle abusive language are primarily reactive and intervene only after abuse has occurred. A complementary approach is developing *proactive* technologies that prevent the harm from occurring in the first place, and we motivate three proactive computational approaches to prevent abuse here.

First, bystanders can have a profound effect on the course of an interaction by steering the direction of the conversation away from abuse (Markey, 2000; Dillon and Bushman, 2015). Prior work has used experimenter-based intervention but a substantial opportunity exists to operationalize these interventions through computational means. Munger (2017) developed a simple, but effective, computational intervention for the use of toxic language (the *n-word*), where a human-looking

bot account would reply with a fixed comment about the harm such language caused and an appeal to empathy, leading to long-term behavior change in the offenders. Identifying how to best respond to abusive behavior—or whether to respond at all—are important computational next steps for this NLP strategy and one that likely needs to be done in collaboration with researchers from fields such as Psychology. Prior work has shown counter speech to be effective for limiting the effects of hate speech (Schieb and Preuss, 2016; Mathew et al., 2018; Stroud and Cox, 2018). Wright et al. (2017) notes that real-world examples of bystanders intervening can be found online, thereby providing a potential source of training data but methods are needed to reliably identify such counter speech examples.

Second, interventions that occur after a point of escalation may have little positive effect in some circumstances. For example, when two individuals have already begun insulting one another, both have already become upset and must lose face to reconcile (Rubin et al., 1994). At this point, de-escalation may prevent further abuse but does little for restoring the situation to a constructive dialog (Gottman, 1999). However, interventions that occur *before* the point of abuse can serve to shift the conversation. Recent work has shown that it is possible to predict whether a conversation will become toxic on Wikipedia (Zhang et al., 2018) and whether bullying will occur on Instagram (Liu et al., 2018). These predictable abuse trajectories open the door to developing new models for pre-emptive interventions that directly mitigate harm.

Third, messages that are not intended as offensive create opportunities to nudge authors towards correcting their text if the offense is pointed out. This strategy builds upon recent work on explainable ML for identifying which parts of a message are offensive (Carton et al., 2018; Noever, 2018), and work on paraphrase and style transfer for suggesting an appropriate inoffensive alternative (Santos et al., 2018; Prabhumoye et al., 2018). For example, parts of a message could be paraphrased to adjust the level of politeness in order to minimize any cumulative disparity towards one social group (Sennrich et al., 2016).

4 Justice Frameworks for NLP

Martin Luther King Jr. wrote that the biggest obstacle to Black freedom is the “white moderate,

who is more devoted to ‘order’ than to justice, who prefers a negative peace which is the absence of tension to a positive peace which is the presence of justice” (King, 1963). Analogously, by focusing only on classifying individual unacceptable speech acts, NLP risks being the same kind of obstacle as the white moderate: Instead of seeking the absence of certain types of speech, we should seek the presence of equitable participation. We argue that NLP should consider supporting three types of justice—social justice, restorative justice, and procedural justice—that describe (i) what actions are allowed and encouraged, (ii) how wrongdoing should be handled, and (iii) what procedures should be followed.

First, the *capabilities approach to social justice* focuses on what actions people can do within a social setting (Sen, 2011; Nussbaum, 2003) and provides a useful framework for thinking about what justice online could look like. Nussbaum (2003) provides a set of 10 fundamental capabilities for a just society, such as the ability to express emotion and to have an affiliation. These capabilities provide a blueprint for articulating the values and opportunities an online community provides: Instead of a negative articulation—an ever-growing list of prohibited behaviors—we should use a positive phrasing (e.g., “you will be able to”) of capabilities in an online community. Such effort naturally extends our proposal for detecting community-specific abuse to one of promoting community norms. Accordingly, NLP technologies can be developed to identify positive behaviors and ensure individuals are able to fulfill these capabilities. Several recent works have made strides in this direction by examining positive behaviors such as how constructive conversations are (Kolhatkar and Taboada, 2017; Napoles et al., 2017), whether dialog on contentious topics can exist without devolving into squabbling (Tan et al., 2016), or the level of support given between community members (Wang and Jurgens, 2018).

Second, once we have adequately articulated what people in a community should be able to do, we must address how the community handles transgressions. The notion of *restorative justice* is a useful theoretical tool for thinking about how wrongdoing should be handled. Restorative justice theory emphasizes repair and uses a process in which stakeholders, including victims and transgressors, decide together on consequences.

A restorative process may produce a punishment, such as banning, but can include consequences such as apology and reconciliation (Braithwaite, 2002). Just responses consider the emotions of both perpetrators and victims in designing the right response (Sherman, 2003). A key problem here is identifying which community norm is violated and NLP technologies can be introduced to aid this process of elucidating violations through classification or use of explainable ML techniques. Here, NLP can aid all parties (platforms, victims, and transgressors) in identifying appropriate avenues for restorative actions.

Third, just communities also require just means of addressing wrongdoing. The notion of *procedural justice* explains that people are more likely to comply with a community’s rules if they believe the authorities are legitimate (Tyler and Huo, 2002; Sherman, 2003). For NLP, it means that our systems for detecting non-compliance must be transparent and fair. People will comply only if they accept the legitimacy of both the platform and the algorithms it employs. Therefore, abuse detection methods are needed to justify why a particular act was a violation to build legitimacy; a natural starting point for NLP in building legitimacy is recent work from explainable ML (Ribeiro et al., 2016; Lei et al., 2016; Carton et al., 2018).

5 Conclusion

Abusive behavior online affects a substantial amount of the population. The NLP community has proposed computational methods to help mitigate this problem, yet has also struggled to move beyond the most obvious tasks in abuse detection. Here, we propose a new strategy for NLP to tackling online abuse in three ways. First, expanding our purview for abuse detection to include both extreme behaviors and the more subtle—but still offensive—behaviors like microaggressions and condescension. Second, NLP must develop methods that go beyond reactive identify-and-delete strategies to one of proactivity that intervenes or nudges individuals to discourage harm *before it occurs*. Third, the community should contextualize its effort inside a broader framework of justice—explicit capabilities, restorative justice, and procedural justice—to directly support the end goal of productive online communities.

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